

Lab: Symmetric Encryption Using OpenSSL

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Step 1: Understanding Symmetric Encryption

Symmetric encryption uses the same key for encryption and decryption. Common symmetric encryption algorithms include:

- AES (Advanced Encryption Standard)
- DES (Data Encryption Standard)
- Triple DES (3DES)
- Blowfish

Step 2: Creating a Secret Message

1. Open a terminal and create a text file:

```
The safe password is 37se86tf
```

```
root@JackLinux:~# echo "The safe password is 37se86tf"
The safe password is 37se86tf
root@JackLinux:~# echo "The safe password is 37se86tf" > secret_message.txt
root@JackLinux:~#
```

2. Verify the content:

```
root@JackLinux:~# cat secret_message.txt
The safe password is 37se86tf
```

Step 3: Encrypting the Message Using AES-256-CBC

1. Use OpenSSL to encrypt the file:

```
root@JackLinux:~# openssl enc -aes-256-cbc -salt -in secret_message.txt -out encrypted_message.enc -pass pass:MySecureKey123!
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

2. Verify the encrypted file exists:

```
root@JackLinux:~# ls -l encrypted_message.enc
-rw-r--r-- 1 root root 48 Sep 21 20:01 encrypted_message.enc
```

Step 4: Decrypting the Message

1. Decrypt the file:

```
root@JackLinux:~# openssl enc -aes-256-cbc -d -in encrypted_message.enc -out decrypted_message.txt -pass pass:MySecureKey123!  
*** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.  
root@JackLinux:~#
```

2. Verify decryption:

```
root@JackLinux:~# cat decrypted_message.txt  
The safe password is 37se86tf  
root@JackLinux:~#
```

Step 5: Experimenting with Different Algorithms

1. Encrypt the file using Blowfish:

```
root@JackLinux:~# openssl enc -bf -salt -in secret_message.txt -out encrypted_blowfish.enc -pass pass:MySecureKey123!  
*** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.
```

2. Decrypt the file:

```
root@JackLinux:~# openssl enc -bf -d -in encrypted_blowfish.enc -out decrypted_blowfish.txt -pass pass:MySecureKey123!  
*** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.
```

3. Verify decryption:

```
cat decrypted_blowfish.txt  
The safe password is 37se86tf
```

Step 6: Cracking Weak Encryption

1. Encrypt the file using DES (AES) (which is now considered weak):

```
root@JackLinux:~# openssl enc -aes-256-cbc -salt -in secret_message.txt -out encrypted_aes.enc -pass pass:123456  
enc: Use -help for summary.  
root@JackLinux:~#
```

2. Attempt brute-force attacks:

I couldn't conduct a brute force attack because the file wouldn't let me encrypt with DES. It was insecure and outdated and even by enabling legacy encryption methods it wouldn't let me use it.